

RELATÓRIO FINAL DETALHADO

Módulos 08 e 09 – Docker

Projeto SmartCall

Documentação técnica detalhada das atividades realizadas na trilha Docker, contemplando gerenciamento e publicação de imagens.

Objetivo Geral: Demonstrar a aplicação prática dos conceitos de imagens Docker, build, camadas, versionamento e gerenciamento seguro no projeto SmartCall.

Evidência 01 – Listagem de imagens

Objetivo: Listar as imagens disponíveis no ambiente local.

Comando executado: docker image ls

Explicação do comando: Exibe todas as imagens armazenadas localmente.

Análise técnica: Foram identificadas imagens oficiais (MySQL e Nginx) e a imagem customizada do SmartCall.

Conclusão: Comprovou o gerenciamento local de imagens Docker.

Captura de tela:

```
● PS C:\Compartilhados\Projetos\SmartCall> docker image ls
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
mysql:8.0	7dcddc01f13b	1.1GB	249MB	U
nginx:latest	5aca99593157	241MB	66MB	U
smartcall-smartcall-app:latest	ccb79f5c8d96	1.76GB	460MB	U

```
○ PS C:\Compartilhados\Projetos\SmartCall> █
```

Evidência 02 – Inspeção da imagem

Objetivo: Inspecionar metadados internos da imagem.

Comando executado: `docker image inspect smartcall-smartcall-app`

Explicação do comando: Exibe arquitetura, variáveis, portas e camadas.

Análise técnica: Foram identificados Linux, amd64, porta 5000 e diretório /app, comprovando a configuração da aplicação.

Conclusão: Validou a estrutura interna da imagem Docker.

Captura de tela:

```
PS C:\Compartilhados\Projetos\SmartCall> docker image inspect smartcall-smartcall-app
[
  {
    "Id": "sha256:ccb79f5c8d9639ccf3e41da499bdf91881fefc3df43fdab7d67ac5ff92bbfa8a",
    "RepoTags": [
      "smartcall-smartcall-app:latest"
    ],
    "RepoDigests": [
      "smartcall-smartcall-app@sha256:ccb79f5c8d9639ccf3e41da499bdf91881fefc3df43fdab7d67ac5ff92bbfa8a"
    ],
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2026-06-08T23:26:01.806490773Z",
    "Config": {
      "ExposedPorts": {
        "5000/tcp": {}
      },
      "Env": [
        "PATH=/usr/local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
        "LANG=C.UTF-8",
        "GPG_KEY=A035C8C19219BA821ECEA86B64E628F8D684696D",
        "PYTHON_VERSION=3.11.15",
        "PYTHON_SHA256=272179ddd9a2e41a0fc8e42e33dfbdca0b3711aa5abf372d3f2d51543d09b625"
      ],
      "Cmd": [
        "python",
        "run.py"
      ],
      "WorkingDir": "/app",
      "Labels": {
        "com.docker.compose.project": "smartcall",
        "com.docker.compose.service": "smartcall-app",
        "com.docker.compose.version": "5.1.3"
      },
      "ArgsEscaped": true
    },
    "Architecture": "amd64",
    "Os": "linux",
    "Size": 459567304,
    "RootFS": {
      "Type": "layers",
      "Layers": [
        "sha256:17d38572a7dcb03eff5b5fd7354c717d7ee9c69b9d6a29f523722534201e411f4",
        "sha256:47dffecc554065cd728ca9db016fb3b7f3b5577dc8964ac34256e3869ada8db9",
        "sha256:8dc335c5db2d262d9c7a006c747d6102580d3797a3d3fc196b3690e61ba53022",
        "sha256:0e6530456c980dd1b447066f1cb9c2b28e32b3ddde18b48ee25491efb38f0382",
        "sha256:e72e1c60729b164dd65c765d8f2f280db4f2d82dbc9654ea42a1d0cc0ba10ef0",
        "sha256:2cdc9b740dea2d0e560142e770209ccfc8b61ec6729ebc99af3a363ae934bc88",
        "sha256:796ab2e16ad279215e13cc1a8de6ebad795632588975129a917d7367de3a376b",
        "sha256:9ce89dfba79cbc3ec638e6588b2bef09194affe1eb7b08799664bfd2629f2beb",
        "sha256:ef6b5fa00a47aa3ee8e58ca6b70f3fc3dfd5431156316801e4946fc50a156260",
        "sha256:b0111c780a0fc84552748f034f4bcbeb0ad657056dc409c18907d0d9503e4042",
        "sha256:b9062864592cce0f8f1b26ac9b0848a7bd40dfd732884c9439ce12870452b63"
      ]
    },
    "Metadata": {
```

Evidência 02 – Continuação

Objetivo: Analisar metadados adicionais.

Comando executado: continuação do inspect

Explicação do comando: Exibe informações de build e digest.

Análise técnica: Os metadados confirmam a rastreabilidade e integridade da imagem.

Conclusão: Complementa a análise da imagem.

Captura de tela:

```
    "Metadata": {
      "LastTagTime": "2026-06-08T23:26:02.109737592Z"
    },
    "Descriptor": {
      "mediaType": "application/vnd.oci.image.index.v1+json",
      "digest": "sha256:ccb79f5c8d9639ccf3e41da499bdf91881fefc3df43fdab7d67ac5ff92bbfa8a",
      "size": 856
    },
    "Identity": {
      "Build": [
        {
          "Ref": "kr32b8qb54cjeolp3fwd5gwo5",
          "CreatedAt": "2026-06-08T23:26:02.296091066Z"
        }
      ]
    }
  ]
}
PS C:\Compartilhados\Projetos\SmartCall>
```

Evidência 03 – Build da imagem

Objetivo: Construir a imagem a partir do Dockerfile.

Comando executado: docker compose -f docker-compose.dev.yml build

Explicação do comando: Executa as etapas definidas no Dockerfile.

Análise técnica: Foi possível observar download da imagem base, camadas, instalação de dependências e exportação da imagem final. O processo evidencia o uso de layers e cache do Docker, reduzindo tempo de builds futuros.

Conclusão: Comprovou a criação da imagem do SmartCall.

Captura de tela:

```
PS C:\Compartilhados\Projetos\SmartCall> docker compose -f docker-compose.dev.yml build
#1 [internal] load local bake definitions
#1 reading from stdin 563B 0.0s done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 201B 0.0s done
#2 DONE 0.0s

#3 [internal] load metadata for docker.io/library/python:3.11
#3 ...

#4 [auth] library/python:pull token for registry-1.docker.io
#4 DONE 0.0s

#3 [internal] load metadata for docker.io/library/python:3.11
#3 DONE 1.9s

#5 [internal] load .dockerignore
#5 transferring context: 2B done
#5 DONE 0.0s

#6 [1/5] FROM docker.io/library/python:3.11@sha256:69feacea6aaa46204e67ee127b30ed727b6c7f7600503cdc1fd1fc0483c16b1c
#6 resolve docker.io/library/python:3.11@sha256:69feacea6aaa46204e67ee127b30ed727b6c7f7600503cdc1fd1fc0483c16b1c 0.0s done
#6 DONE 0.2s

#7 [internal] load build context
#7 transferring context: 83.21kB 0.2s done
#7 DONE 0.2s

#6 [1/5] FROM docker.io/library/python:3.11@sha256:69feacea6aaa46204e67ee127b30ed727b6c7f7600503cdc1fd1fc0483c16b1c
#6 sha256:00fb515ecb79dbdde9a261142091767790a778357d0288935f246d7715fb5286 0B / 23.98MB 0.2s
#6 sha256:00fb515ecb79dbdde9a261142091767790a778357d0288935f246d7715fb5286 2.10MB / 23.98MB 0.3s
#6 sha256:e5c164e2c953103f6ad6a958675f1cbe93636f151e54766f977bb8c0eeab4ba4 0B / 249B 0.2s
#6 sha256:f6fb30da281d5d9a78bb2970aeb5c66a90168f54bb092ad3af180dba6e029fe1 0B / 6.09MB 0.2s
#6 sha256:442cca122a8b0987d6ea29e5fa7d0b8caf699cb127ec5fdbd8191faf86da521f 0B / 236.20MB 0.2s
#6 sha256:00fb515ecb79dbdde9a261142091767790a778357d0288935f246d7715fb5286 4.19MB / 23.98MB 0.5s
#6 sha256:00fb515ecb79dbdde9a261142091767790a778357d0288935f246d7715fb5286 6.29MB / 23.98MB 0.6s
#6 sha256:00fb515ecb79dbdde9a261142091767790a778357d0288935f246d7715fb5286 9.44MB / 23.98MB 0.8s
#6 sha256:00fb515ecb79dbdde9a261142091767790a778357d0288935f246d7715fb5286 11.53MB / 23.98MB 0.9s
#6 sha256:e5c164e2c953103f6ad6a958675f1cbe93636f151e54766f977bb8c0eeab4ba4 249B / 249B 0.8s done
#6 sha256:f6fb30da281d5d9a78bb2970aeb5c66a90168f54bb092ad3af180dba6e029fe1 1.05MB / 6.09MB 0.8s
#6 sha256:f6fb30da281d5d9a78bb2970aeb5c66a90168f54bb092ad3af180dba6e029fe1 3.15MB / 6.09MB 0.9s
#6 sha256:2ca884014342f240be01d391b498c3ec61b2f4af2c205e7e9a0b5ac2cb2f24c4 0B / 67.78MB 0.2s
#6 sha256:00fb515ecb79dbdde9a261142091767790a778357d0288935f246d7715fb5286 14.68MB / 23.98MB 1.2s
#6 sha256:f6fb30da281d5d9a78bb2970aeb5c66a90168f54bb092ad3af180dba6e029fe1 4.19MB / 6.09MB 1.1s
#6 sha256:f6fb30da281d5d9a78bb2970aeb5c66a90168f54bb092ad3af180dba6e029fe1 6.09MB / 6.09MB 1.3s done
#6 sha256:00fb515ecb79dbdde9a261142091767790a778357d0288935f246d7715fb5286 16.78MB / 23.98MB 1.4s
#6 sha256:00fb515ecb79dbdde9a261142091767790a778357d0288935f246d7715fb5286 18.87MB / 23.98MB 1.5s
#6 sha256:8b365b6ff7b7e310e1b9a88f996e65b89969c7fa450492b68249d1d1c38e0676 0B / 25.64MB 0.2s
#6 sha256:00fb515ecb79dbdde9a261142091767790a778357d0288935f246d7715fb5286 22.02MB / 23.98MB 1.8s
#6 sha256:8b365b6ff7b7e310e1b9a88f996e65b89969c7fa450492b68249d1d1c38e0676 2.10MB / 25.64MB 0.6s
#6 sha256:00fb515ecb79dbdde9a261142091767790a778357d0288935f246d7715fb5286 23.98MB / 23.98MB 2.0s done
#6 sha256:ef1b08dd59d42b444e8f9c68a6cebbcd70636aed0c242fddcc0bd61b61ba26b 0B / 49.32MB 0.2s
#6 sha256:8b365b6ff7b7e310e1b9a88f996e65b89969c7fa450492b68249d1d1c38e0676 4.19MB / 25.64MB 0.9s
```

Evidência 03 – Continuação

Objetivo: Acompanhar o processo de build.

Comando executado: build

Explicação do comando: Extração de camadas.

Análise técnica: Demonstrou o mecanismo de camadas reutilizáveis do Docker.

Conclusão: Reforça o conceito de cache.

Captura de tela:

```
PS C:\Compartilhados\Projetos\SmartCall> docker compose -f docker-compose.dev.yml build
#6 sha256:8b365b6fff7b7e310e1b9a88f996e65b89969c7fa450492b68249d1d1c38e0676 4.19MB / 25.64MB 0.9s
#6 sha256:2ca884014342f240be01d391b498c3ec61b2f4af2c205e7e9a0b5ac2cb2f24c4 4.19MB / 67.78MB 1.8s
#6 sha256:8b365b6fff7b7e310e1b9a88f996e65b89969c7fa450492b68249d1d1c38e0676 6.29MB / 25.64MB 1.4s
#6 sha256:ef1b08ddd59d42b444e8f9c68a6cebbbed70636aed0c242fddcb0bd61b61ba26b 3.15MB / 49.32MB 0.8s
#6 sha256:8b365b6fff7b7e310e1b9a88f996e65b89969c7fa450492b68249d1d1c38e0676 8.39MB / 25.64MB 1.7s
#6 sha256:442cca122a8b0987d6ea29e5fa7d0b8caf699cb127ec5fddb8191faf86da521f 12.58MB / 236.20MB 3.2s
#6 sha256:ef1b08ddd59d42b444e8f9c68a6cebbbed70636aed0c242fddcb0bd61b61ba26b 6.29MB / 49.32MB 1.2s
#6 sha256:8b365b6fff7b7e310e1b9a88f996e65b89969c7fa450492b68249d1d1c38e0676 10.49MB / 25.64MB 2.1s
#6 sha256:2ca884014342f240be01d391b498c3ec61b2f4af2c205e7e9a0b5ac2cb2f24c4 8.39MB / 67.78MB 3.0s
#6 sha256:8b365b6fff7b7e310e1b9a88f996e65b89969c7fa450492b68249d1d1c38e0676 12.58MB / 25.64MB 2.6s
#6 sha256:ef1b08ddd59d42b444e8f9c68a6cebbbed70636aed0c242fddcb0bd61b61ba26b 9.44MB / 49.32MB 1.8s
#6 sha256:8b365b6fff7b7e310e1b9a88f996e65b89969c7fa450492b68249d1d1c38e0676 14.68MB / 25.64MB 3.1s
#6 sha256:442cca122a8b0987d6ea29e5fa7d0b8caf699cb127ec5fddb8191faf86da521f 26.21MB / 236.20MB 4.7s
#6 sha256:ef1b08ddd59d42b444e8f9c68a6cebbbed70636aed0c242fddcb0bd61b61ba26b 12.58MB / 49.32MB 2.5s
#6 sha256:8b365b6fff7b7e310e1b9a88f996e65b89969c7fa450492b68249d1d1c38e0676 16.78MB / 25.64MB 3.6s
#6 sha256:2ca884014342f240be01d391b498c3ec61b2f4af2c205e7e9a0b5ac2cb2f24c4 12.58MB / 67.78MB 4.2s
#6 sha256:ef1b08ddd59d42b444e8f9c68a6cebbbed70636aed0c242fddcb0bd61b61ba26b 15.73MB / 49.32MB 3.1s
#6 sha256:8b365b6fff7b7e310e1b9a88f996e65b89969c7fa450492b68249d1d1c38e0676 18.87MB / 25.64MB 4.1s
#6 sha256:442cca122a8b0987d6ea29e5fa7d0b8caf699cb127ec5fddb8191faf86da521f 39.85MB / 236.20MB 5.8s
#6 sha256:2ca884014342f240be01d391b498c3ec61b2f4af2c205e7e9a0b5ac2cb2f24c4 16.78MB / 67.78MB 5.1s
#6 sha256:8b365b6fff7b7e310e1b9a88f996e65b89969c7fa450492b68249d1d1c38e0676 20.97MB / 25.64MB 4.6s
#6 sha256:ef1b08ddd59d42b444e8f9c68a6cebbbed70636aed0c242fddcb0bd61b61ba26b 18.87MB / 49.32MB 4.1s
#6 sha256:8b365b6fff7b7e310e1b9a88f996e65b89969c7fa450492b68249d1d1c38e0676 23.07MB / 25.64MB 5.1s
#6 sha256:442cca122a8b0987d6ea29e5fa7d0b8caf699cb127ec5fddb8191faf86da521f 52.43MB / 236.20MB 6.9s
#6 sha256:8b365b6fff7b7e310e1b9a88f996e65b89969c7fa450492b68249d1d1c38e0676 25.64MB / 25.64MB 5.7s done
#6 sha256:ef1b08ddd59d42b444e8f9c68a6cebbbed70636aed0c242fddcb0bd61b61ba26b 22.02MB / 49.32MB 4.1s
#6 sha256:2ca884014342f240be01d391b498c3ec61b2f4af2c205e7e9a0b5ac2cb2f24c4 20.97MB / 67.78MB 6.3s
#6 sha256:442cca122a8b0987d6ea29e5fa7d0b8caf699cb127ec5fddb8191faf86da521f 66.06MB / 236.20MB 7.8s
#6 sha256:ef1b08ddd59d42b444e8f9c68a6cebbbed70636aed0c242fddcb0bd61b61ba26b 25.17MB / 49.32MB 5.7s
#6 sha256:2ca884014342f240be01d391b498c3ec61b2f4af2c205e7e9a0b5ac2cb2f24c4 25.17MB / 67.78MB 7.3s
#6 sha256:442cca122a8b0987d6ea29e5fa7d0b8caf699cb127ec5fddb8191faf86da521f 79.69MB / 236.20MB 8.7s
#6 sha256:ef1b08ddd59d42b444e8f9c68a6cebbbed70636aed0c242fddcb0bd61b61ba26b 28.31MB / 49.32MB 6.6s
#6 sha256:442cca122a8b0987d6ea29e5fa7d0b8caf699cb127ec5fddb8191faf86da521f 92.27MB / 236.20MB 9.4s
#6 sha256:2ca884014342f240be01d391b498c3ec61b2f4af2c205e7e9a0b5ac2cb2f24c4 29.36MB / 67.78MB 8.7s
#6 sha256:ef1b08ddd59d42b444e8f9c68a6cebbbed70636aed0c242fddcb0bd61b61ba26b 31.46MB / 49.32MB 7.5s
#6 sha256:442cca122a8b0987d6ea29e5fa7d0b8caf699cb127ec5fddb8191faf86da521f 104.86MB / 236.20MB 10.2s
#6 sha256:ef1b08ddd59d42b444e8f9c68a6cebbbed70636aed0c242fddcb0bd61b61ba26b 34.60MB / 49.32MB 8.2s
#6 sha256:2ca884014342f240be01d391b498c3ec61b2f4af2c205e7e9a0b5ac2cb2f24c4 33.55MB / 67.78MB 9.7s
#6 sha256:442cca122a8b0987d6ea29e5fa7d0b8caf699cb127ec5fddb8191faf86da521f 119.54MB / 236.20MB 11.1s
#6 sha256:ef1b08ddd59d42b444e8f9c68a6cebbbed70636aed0c242fddcb0bd61b61ba26b 37.75MB / 49.32MB 9.1s
#6 sha256:2ca884014342f240be01d391b498c3ec61b2f4af2c205e7e9a0b5ac2cb2f24c4 37.75MB / 67.78MB 10.9s
#6 sha256:ef1b08ddd59d42b444e8f9c68a6cebbbed70636aed0c242fddcb0bd61b61ba26b 40.89MB / 49.32MB 9.7s
#6 sha256:8b365b6fff7b7e310e1b9a88f996e65b89969c7fa450492b68249d1d1c38e0676 44.04MB / 49.32MB 10.2s
#6 sha256:442cca122a8b0987d6ea29e5fa7d0b8caf699cb127ec5fddb8191faf86da521f 132.12MB / 236.20MB 12.4s
#6 sha256:2ca884014342f240be01d391b498c3ec61b2f4af2c205e7e9a0b5ac2cb2f24c4 41.94MB / 67.78MB 11.6s
#6 sha256:ef1b08ddd59d42b444e8f9c68a6cebbbed70636aed0c242fddcb0bd61b61ba26b 47.19MB / 49.32MB 10.5s
#6 sha256:8b365b6fff7b7e310e1b9a88f996e65b89969c7fa450492b68249d1d1c38e0676 49.32MB / 49.32MB 10.8s done
#6 sha256:2ca884014342f240be01d391b498c3ec61b2f4af2c205e7e9a0b5ac2cb2f24c4 46.14MB / 67.78MB 12.0s
#6 extracting sha256:ef1b08ddd59d42b444e8f9c68a6cebbbed70636aed0c242fddcb0bd61b61ba26b
#6 sha256:2ca884014342f240be01d391b498c3ec61b2f4af2c205e7e9a0b5ac2cb2f24c4 50.33MB / 67.78MB 12.5s
#6 sha256:442cca122a8b0987d6ea29e5fa7d0b8caf699cb127ec5fddb8191faf86da521f 144.70MB / 236.20MB 13.6s
#6 sha256:2ca884014342f240be01d391b498c3ec61b2f4af2c205e7e9a0b5ac2cb2f24c4 54.53MB / 67.78MB 12.9s
#6 sha256:2ca884014342f240be01d391b498c3ec61b2f4af2c205e7e9a0b5ac2cb2f24c4 59.77MB / 67.78MB 13.3s
#6 sha256:2ca884014342f240be01d391b498c3ec61b2f4af2c205e7e9a0b5ac2cb2f24c4 65.01MB / 67.78MB 13.8s
```


Evidência 03 – Continuação

Objetivo: Acompanhar execução do Dockerfile.

Comando executado: build

Explicação do comando: Executa WORKDIR, COPY e RUN.

Análise técnica: Comprova a automação do ambiente da aplicação.

Conclusão: Valida o Dockerfile.

Captura de tela:

```
PS C:\Compartilhados\Projetos\SmartCall> docker compose -f docker-compose.dev.yml build
#6 sha256:2ca884014342f240be01d391b498c3ec61b2f4af2c205e7e9a0b5ac2cb2f24c4 65.01MB / 67.78MB 13.8s
#6 sha256:2ca884014342f240be01d391b498c3ec61b2f4af2c205e7e9a0b5ac2cb2f24c4 67.78MB / 67.78MB 14.0s done
#6 sha256:442cca122a8b0987d6ea29e5fa7d0b8caf699cb127ec5fdbd8191faf86da521f 161.48MB / 236.20MB 15.0s
#6 sha256:442cca122a8b0987d6ea29e5fa7d0b8caf699cb127ec5fdbd8191faf86da521f 175.11MB / 236.20MB 15.7s
#6 sha256:442cca122a8b0987d6ea29e5fa7d0b8caf699cb127ec5fdbd8191faf86da521f 187.70MB / 236.20MB 16.3s
#6 extracting sha256:ef1b08ddd59d42b444e8f9c68a6cebbbed70636aed0c242fdccb0bd61b61ba26b 3.6s done
#6 extracting sha256:8b365b6ff7b7e310e1b9a88f996e65b89969c7fa450492b68249d1d1c38e0676
#6 sha256:442cca122a8b0987d6ea29e5fa7d0b8caf699cb127ec5fdbd8191faf86da521f 200.28MB / 236.20MB 17.1s
#6 sha256:442cca122a8b0987d6ea29e5fa7d0b8caf699cb127ec5fdbd8191faf86da521f 212.86MB / 236.20MB 17.7s
#6 sha256:442cca122a8b0987d6ea29e5fa7d0b8caf699cb127ec5fdbd8191faf86da521f 225.44MB / 236.20MB 18.3s
#6 sha256:442cca122a8b0987d6ea29e5fa7d0b8caf699cb127ec5fdbd8191faf86da521f 236.20MB / 236.20MB 20.0s done
#6 extracting sha256:8b365b6ff7b7e310e1b9a88f996e65b89969c7fa450492b68249d1d1c38e0676 3.6s done
#6 extracting sha256:2ca884014342f240be01d391b498c3ec61b2f4af2c205e7e9a0b5ac2cb2f24c4
#6 extracting sha256:2ca884014342f240be01d391b498c3ec61b2f4af2c205e7e9a0b5ac2cb2f24c4 6.4s done
#6 extracting sha256:442cca122a8b0987d6ea29e5fa7d0b8caf699cb127ec5fdbd8191faf86da521f
#6 extracting sha256:442cca122a8b0987d6ea29e5fa7d0b8caf699cb127ec5fdbd8191faf86da521f 15.2s done
#6 DONE 41.9s

#6 [1/5] FROM docker.io/library/python:3.11@sha256:69feacea6aaa46204e67ee127b30ed727b6c7f7600503cdc1fd1fc0483c16b1c
#6 extracting sha256:f6fb30da281d5d9a78bb2970aeb5c66a90168f54bb092ad3af180dba6e029fe1
#6 extracting sha256:f6fb30da281d5d9a78bb2970aeb5c66a90168f54bb092ad3af180dba6e029fe1 0.4s done
#6 extracting sha256:00fb515ecb79dbdde9a261142091767790a778357d0288935f246d7715fb5286
#6 extracting sha256:00fb515ecb79dbdde9a261142091767790a778357d0288935f246d7715fb5286 1.7s done
#6 DONE 44.0s

#6 [1/5] FROM docker.io/library/python:3.11@sha256:69feacea6aaa46204e67ee127b30ed727b6c7f7600503cdc1fd1fc0483c16b1c
#6 extracting sha256:e5c164e2c953103f6ad6a958675f1cbe93636f151e54766f977bb8c0eeab4ba4 0.0s done
#6 DONE 44.0s

#8 [2/5] WORKDIR /app
#8 DONE 1.4s

#9 [3/5] COPY requirements.txt .
#9 DONE 0.1s

#10 [4/5] RUN pip install flask mysql-connector-python
#10 1.920 Collecting flask
#10 2.378 Downloading flask-3.1.3-py3-none-any.whl.metadata (3.2 kB)
#10 2.467 Collecting mysql-connector-python
#10 2.483 Downloading mysql_connector_python-9.7.0-cp311-cp311-manylinux_2_28_x86_64.whl.metadata (11 kB)
#10 2.537 Collecting blinker>=1.9.0 (from flask)
#10 2.553 Downloading blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
#10 2.609 Collecting click>=8.1.3 (from flask)
#10 2.625 Downloading click-8.4.1-py3-none-any.whl.metadata (2.6 kB)
#10 2.674 Collecting itsdangerous>=2.2.0 (from flask)
#10 2.689 Downloading itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
#10 2.741 Collecting jinja2>=3.1.2 (from flask)
#10 2.756 Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
#10 2.854 Collecting markupsafe>=2.1.1 (from flask)
#10 2.869 Downloading markupsafe-3.0.3-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (2.7 kB)
#10 2.934 Collecting werkzeug>=3.1.0 (from flask)
#10 2.949 Downloading werkzeug-3.1.8-py3-none-any.whl.metadata (4.0 kB)
#10 2.992 Downloading flask-3.1.3-py3-none-any.whl (103 kB)
#10 3.020 _____ 103.4/103.4 kB 3.8 MB/s eta 0:00:00
```

Evidência 03 – Finalização

Objetivo: Concluir a geração da imagem.

Comando executado: build

Explicação do comando: Exporta e registra a imagem.

Análise técnica: A imagem smartcall-smartcall-app foi construída com sucesso.

Conclusão: Build concluído.

Captura de tela:

```
#10 2.992 Downloading flask-3.1.3-py3-none-any.whl (103 kB)
#10 3.020 _____ 103.4/103.4 kB 3.8 MB/s eta 0:00:00
#10 3.040 Downloading mysql_connector_python-9.7.0-cp311-cp311-manylinux_2_28_x86_64.whl (21.7 MB)
#10 5.398 _____ 21.7/21.7 MB 6.3 MB/s eta 0:00:00
#10 5.412 Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
#10 5.439 Downloading click-8.4.1-py3-none-any.whl (116 kB)
#10 5.458 _____ 116.6/116.6 kB 6.5 MB/s eta 0:00:00
#10 5.472 Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
#10 5.497 Downloading Jinja2-3.1.6-py3-none-any.whl (134 kB)
#10 5.517 _____ 134.9/134.9 kB 7.7 MB/s eta 0:00:00
#10 5.531 Downloading markupsafe-3.0.3-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (22 kB)
#10 5.558 Downloading werkzeug-3.1.0-py3-none-any.whl (226 kB)
#10 5.584 _____ 226.5/226.5 kB 10.0 MB/s eta 0:00:00
#10 5.659 Installing collected packages: mysql-connector-python, markupsafe, itsdangerous, click, blinker, werkzeug, Jinja2, flask
#10 6.566 Successfully installed blinker-1.9.0 click-8.4.1 flask-3.1.3 itsdangerous-2.2.0 Jinja2-3.1.6 markupsafe-3.0.3 mysql-connector-python-9.7.0 werkzeug-3.1.0
#10 6.566 WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv
#10 6.719
#10 6.719 [notice] A new release of pip is available: 24.0 -> 26.1.2
#10 6.719 [notice] To update, run: pip install --upgrade pip
#10 DONE 7.0s

#11 [5/5] COPY . .
#11 DONE 0.3s

#12 exporting to image
#12 exporting layers
#12 exporting layers 3.7s done
#12 exporting manifest sha256:5a893d8290449ad8760744b1f71c1219ec99a47083434d78c3d00f5f705eb658 0.0s done
#12 exporting config sha256:788625c5fbd4b91ba8cf4d28f174dde2b1f22c9f5b2182a85e1eb7e01b730bc5 0.0s done
#12 exporting attestation manifest sha256:f832006e7956a09362f6cceb35ab209a70e7e8a908ef11f80f3d628a3728ec56 0.0s done
#12 exporting manifest list sha256:a2ca69f2f456089210bf924f0e801ab4219bb382dce48775c2e09a6ee4755
#12 exporting manifest list sha256:a2ca69f2f456089210bf924f0e801ab4219bb382dce48775c2e09a6ee4755 0.0s done
#12 naming to docker.io/library/smartcall-smartcall-app:latest done
#12 unpacking to docker.io/library/smartcall-smartcall-app:latest
#12 unpacking to docker.io/library/smartcall-smartcall-app:latest 3.0s done
#12 DONE 6.9s

#13 resolving provenance for metadata file
#13 DONE 0.0s
[+] build 1/1
✓ Image smartcall-smartcall-app Built
ps C:\Compartilhados\Projetos\SmartCall>
```

Evidência 04 – Imagens após build

Objetivo: Validar a presença da imagem após build.

Comando executado: docker image ls

Explicação do comando: Lista imagens locais.

Análise técnica: A imagem permaneceu disponível no ambiente.

Conclusão: Confirma persistência da imagem.

Captura de tela:

```
● PS C:\Compartilhados\Projetos\SmartCall> docker image ls
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
mysql:8.0	7dcddc01f13b	1.1GB	249MB	U
nginx:latest	5aca99593157	241MB	66MB	U
smartcall-smartcall-app:latest	aa2ca69f2fa5	1.76GB	460MB	

Evidência 05 – Versionamento

Objetivo: Criar versão da imagem.

Comando executado: `docker tag ...:1.0`

Explicação do comando: Adiciona uma nova tag à imagem.

Análise técnica: A imagem passou a possuir as tags latest e 1.0, prática comum em CI/CD.

Conclusão: Comprovou versionamento.

Captura de tela:

```
PS C:\Compartilhados\Projetos\SmartCall> docker image ls
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
mysql:8.0	7dcddc01f13b	1.1GB	249MB	U
nginx:latest	5aca99593157	241MB	66MB	U
smartcall-smartcall-app:latest	aa2ca69f2fa5	1.76GB	460MB	

```
PS C:\Compartilhados\Projetos\SmartCall> docker tag smartcall-smartcall-app:latest smartcall-smartcall-app:1.0
PS C:\Compartilhados\Projetos\SmartCall> docker image ls
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
mysql:8.0	7dcddc01f13b	1.1GB	249MB	U
nginx:latest	5aca99593157	241MB	66MB	U
smartcall-smartcall-app:1.0	aa2ca69f2fa5	1.76GB	460MB	
smartcall-smartcall-app:latest	aa2ca69f2fa5	1.76GB	460MB	

```
PS C:\Compartilhados\Projetos\SmartCall>
```

Evidência 06 – Imagens órfãs

Objetivo: Verificar imagens não utilizadas.

Comando executado: `docker image ls -f dangling=true`

Explicação do comando: Lista imagens sem tag.

Análise técnica: Nenhuma imagem órfã foi encontrada, indicando ambiente organizado.

Conclusão: Comprovou boas práticas.

Captura de tela:

```
PS C:\Compartilhados\Projetos\SmartCall> docker image ls -f dangling=true
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
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Evidência 07 – Contêineres x Imagens

Objetivo: Mapear imagens em uso.

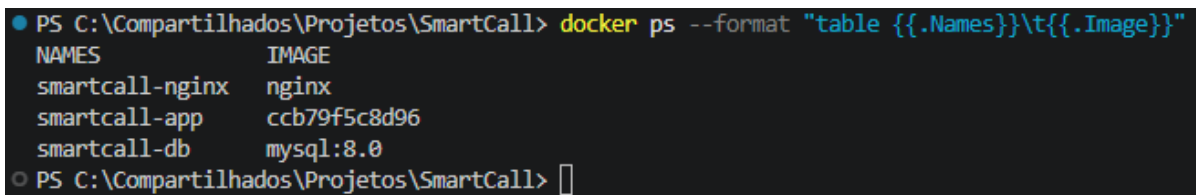
Comando executado: `docker ps --format "table {{.Names}}\t{{.Image}}"`

Explicação do comando: Relaciona contêineres às imagens.

Análise técnica: Foi possível identificar as imagens utilizadas por cada serviço do SmartCall.

Conclusão: Permite administração segura do ambiente.

Captura de tela:



```
PS C:\Compartilhados\Projetos\SmartCall> docker ps --format "table {{.Names}}\t{{.Image}}"
NAMES          IMAGE
smartcall-nginx nginx
smartcall-app   ccb79f5c8d96
smartcall-db    mysql:8.0
PS C:\Compartilhados\Projetos\SmartCall> 
```

Conclusão Geral

O projeto SmartCall atendeu integralmente aos requisitos do Módulo 08, incluindo criação, inspeção, build, versionamento e gerenciamento de imagens Docker. O Módulo 09 foi parcialmente concluído, não contemplando a publicação no Docker Hub por não fazer parte dos requisitos atuais do projeto. As evidências apresentadas demonstram aplicação prática dos conceitos em ambiente corporativo real.